

ELECTROMYOGRAPHIC ANALYSIS OF THE GLUTEUS MEDIUS IN FIVE WEIGHT-BEARING EXERCISES

DAVID A. KRAUSE,¹ REBECCA S. JACOBS,² KATIE E. PILGER,² BECKY R. SATHER,² SETH P. SIBUNKA,² AND JOHN H. HOLLMAN¹

¹Physical Medicine and Rehabilitation, Mayo Clinic, Rochester, Minnesota; and ²Doctor of Physical Therapy Program, Mayo Clinic, Rochester, Minnesota

ABSTRACT

Krause, DA, Jacobs, RS, Pilger, KE, Sather, BR, Sibunka, SP, and Hollman, JH. Electromyographic analysis of the gluteus medius in five weight-bearing exercises. *J Strength Cond Res* 23(9): 2689–2694, 2009—Weight-bearing exercises are frequently used to train and strengthen muscles of the hip. These exercises have been advocated in the rehabilitation of a variety of hip and knee dysfunctions. Limited evidence is available to describe the level of muscle activation occurring with specific weight-bearing exercises. The purpose of this study was to investigate the level of activation of the gluteus medius muscle as measured by electromyographic (EMG) signal amplitude in 5 weight-bearing exercises. Twenty healthy subjects aged 21 to 30 years participated in the study. The EMG surface electrodes were positioned over the muscle belly of the gluteus medius. Subjects performed 5 exercises that consisted of bilateral stance, single limb stance, single limb stance on both a firm surface and an Airex cushion, and single limb squat on a firm surface and an Airex cushion. Statistical differences ($p < 0.05$) in gluteus medius EMG values were found between single limb stance as compared with double limb stance, and single limb squat as compared with single limb stance. Single limb stance places more demands on the gluteus medius than double limb stance, whereas single limb squats are more demanding than single limb stance. Although exercises performed on an Airex cushion produced greater EMG values as compared with a firm surface, the difference was not statistically significant. The results, however, suggest that if the goal is to increase the challenge to the gluteus medius, dynamic, single limb exercises performed on unstable surfaces, such as a balance cushion, may place greater demands on the gluteus medius than similar exercises performed on stable surfaces.

KEY WORDS electromyography, hip, exercise prescription

Address correspondence to David A. Krause, krause.david@mayo.edu.
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INTRODUCTION

Unilateral weight-bearing exercises are often used to train and strengthen muscles of the hip. The gluteus medius, which acts to stabilize the hip in both the frontal and transverse planes, is often the primary muscle of interest in these exercises. If the hip lacks stability during single limb weight-bearing activities, the femur may adduct and internally rotate, leading to increased valgus at the knee and pronation of the foot. This is termed medial collapse of the lower extremity (14) and has been reported to play a role in lower-extremity injuries including anterior cruciate ligament tears (6), iliotibial band syndrome (7), and patellofemoral pain syndrome (8–10, 12, 13). Patellofemoral pain is a common orthopedic complaint among young adults. Factors related to the onset of patellofemoral pain have included those related to the patellofemoral joint itself and factors both distal and proximal to the knee (8, 9, 12–14, 16, 17). Ireland et al. (10), for example, found that females with patellofemoral pain were more likely to demonstrate weakness in hip abduction as well as external rotation than age-matched women without patellofemoral pain.

Electromyography (EMG) is an established way to quantify muscle activity. Previous research has measured EMG activation of the gluteus medius during dynamic and isometric single leg activities in weight-bearing and non-weight-bearing positions (1, 4, 5, 15). One variable that has not been reported is the influence of stable and unstable surfaces while performing weight-bearing bilateral and unilateral hip strengthening exercises. For functional reasons, individuals must be able to adjust for changing surfaces during many daily activities. It is not known whether the muscle activity of the gluteus medius changes in response to stable and unstable surfaces. Although surface changes are frequently encountered in community ambulation, manipulation of surface stability may be used as a component of a rehabilitation program for strengthening the gluteus medius. Examples of rehabilitation programs incorporating weight-bearing exercises on unstable surfaces for neuromuscular training include anterior cruciate ligament preventative and rehabilitative programs and patellofemoral pain programs. As stated previously, hip abductors, including the gluteus medius, are recruited during weight-bearing activities to

maintain proper alignment of the femur and pelvis. Gluteus medius exercises are thus incorporated into lower-extremity preventative and rehabilitative programs given the association of reduced hip abduction strength with lower-extremity malalignments and injuries (9,10,13). One goal of our study was to measure gluteus medius activity during static and dynamic single limb exercise on both stable and unstable surfaces. Understanding the relative differences in EMG activity will assist clinicians in incorporating hip exercises into treatment and prevention programs based on the level of demand to the gluteus medius desired.

The purpose of this study was to investigate the level of activation of the gluteus medius muscle as measured by surface EMG signal amplitude during 5 weight-bearing exercises performed on stable and unstable surfaces. We hypothesized a) single limb exercises require greater gluteus medius activity than double limb stance, b) dynamic exercises require greater gluteus medius activity than static exercises, and c) exercise on an unstable surface will require greater gluteus medius activity than the same exercise done on the firm surface.

METHODS

Experimental Approach to the Problem

In this study, we examined the influence of exercise and support surface stability on the activation level of the gluteus medius. The EMG signals were collected with surface electrodes, processed with a root mean square (RMS) algorithm for all conditions, and normalized to a maximal voluntary isometric contraction (MVIC). A repeated measures analysis of variance (ANOVA) was completed to determine whether there was a statistical significance within the data set. Post hoc Bonferroni adjusted *t*-tests were run to test the hypotheses that a) single limb exercises require greater activation than double limb stance, b) dynamic exercises require greater activation than static activities, and c) exercises on an unstable surface require greater activation than the same exercises done on a firm surface. The 5 exercises, double limb stance on a firm surface, single limb stance on a firm surface, single limb squat on a firm surface, single limb stance on an Airex cushion (Alcon, Inc., Montreal, Canada), and single limb squat on an Airex cushion, were randomized to reduce any order threats to the study's internal validity.

Subjects

Twenty healthy subjects (14 female and 6 male) were recruited to participate in this study. The activity and participation level of the subjects was primarily involvement in recreational sporting and fitness activities. The subjects ranged from 21 to 30 years of age. The average age of females was 23.6 ± 1.7 years, with a mean height of 169.3 ± 9.5 cm, and mean weight of 65.0 ± 9.2 kg. The average of males was 26.3 ± 2.5 years, with a mean height of 172.2 ± 12.9 cm, and a mean weight of 85.0 ± 10.1 kg. Inclusion criteria included

healthy individuals between the ages of 18 and 35. Exclusion criteria included any lower-extremity conditions on subjects' dominant leg limiting the ability to perform hip strengthening exercises. Subjects provided signed informed consent before participation. The study was approved by the Institutional Review Board at the Mayo Clinic, Rochester, MN.

Procedures

Raw EMG signals of the gluteus medius were collected with D-100 bipolar surface electrodes (active Ag-AgCl electrodes with an interelectrode distance of 22 mm encased within preamplifier assemblies measuring $35 \times 17 \times 10$ mm with a gain 35). Electrode leads from the preamplifiers were connected to a GCS67 main amplifier system (Therapeutics Unlimited, Inc., Iowa City, IA, USA). The combined preamplifier and main amplifier permitted a gain of 100 to 10,000 with a bandwidth of 40 Hz to 6 KHz. The common mode rejection ratio was 87 dB at 60 Hz, and input impedance was greater than 15 M Ω at 100 Hz. Data were collected at a sampling frequency of 1,000 Hz. Raw EMG signals were processed with WinDaq data acquisition software (DATAQ Instruments, Inc., Akron, OH, USA).

The EMG recording electrode was placed over the muscle belly of the gluteus medius half the distance between the



Figure 1. Exercise 1: double limb stance on floor. Subjects stood with both feet on floor, shoulder width apart with knees extended. Position was held for 10 seconds and repeated 3 times.

greater trochanter and the iliac crest (14). The ground electrode was placed over the proximal shaft of the tibia. Before placement of the electrodes, the skin was prepared by rubbing an alcohol wipe over the areas the electrodes would be placed. Conduction gel was applied to the surface electrodes before they were applied to the skin. Clear plastic tape was used to secure the electrodes in place.

Subjects walked for 5 minutes as a warm-up exercise before testing. The dominant leg was determined by asking the participants which leg they used to kick a ball. The EMG surface electrodes were then placed on the subject by an examiner.

An MVIC was obtained as a normalization reference in analysis of the EMG activity during the 5 exercise conditions. The MVIC was recorded using a break manual muscle test for the gluteus medius as described by Kendall et al. (11). The patient was positioned sidelying on their nondominant limb. The upper test limb was positioned with the hip slightly extended beyond midline, knee extended, pelvis rotated slightly forward, with the lower limb flexed at the hip and

knee for stability. Standing behind the patient, the examiner used a long lever to perform the break test of the gluteus medius. Resistance was provided just proximal to the ankle. The test was performed with the test limb abducted to 30°. The examiner applied a straight, downward force while the subject attempted to maintain the position of abduction. Three trials were performed with adequate rest between bouts. The EMG data were recorded during the third trial.

A within-subjects design was used with each subject performing the 5 test exercises described in Figures 1 to 5. The order of the exercises was randomized for each subject. All exercises were practiced before the actual randomization and testing process. Each subject performed 3 trials of each exercise. Adequate rest time was provided between each set of the individual exercises to avoid fatigue. For the stance exercise, EMG was recorded for 10 seconds. For the squat exercises, EMG was recorded continuously for a series of 3 squats.



Figure 2. Exercise 2: single limb stance on floor. Subjects stood on 1 foot with hands on hips and knee slightly bent. This position was held for 10 seconds and repeated 3 times.



Figure 3. Exercise 3: single limb squat on floor. Subjects squatted to 45° of knee flexion and then returned to standing. Descent phase duration was 1 second, as was ascent phase. Pacing was monitored with metronome. Subjects performed 3 consecutive squats. Series of 3 squats was repeated 3 times.



Figure 4. Exercise 4: single limb stance on Airex cushion. Same exercise as described for single limb stance on floor but performed on Airex cushion.



Figure 5. Exercise 5: single limb squat on Airex cushion. Same exercise as described for the single limb squat on the floor however performed on an Airex cushion.

To avoid lateral shifting of the upper body during the single limb exercises, a plumb line was positioned in front of the subject in reference to his/her midline. The vertical position of the trunk and head and level of the position of the pelvis was monitored by one of the examiners. If deviation occurred on any repetition, the task was repeated. A second examiner monitored the level of knee flexion from the side of the subject during the exercises. For the stance exercises, the knee was maintained slightly flexed to avoid a locked or hyperextended position. During squat exercises, the subject descended from an extended knee position to 45° of knee flexion. This was also monitored by the second examiner with a handheld goniometer.

Statistical Analyses

Raw EMG signals recorded during the 5 exercises were processed with the RMS algorithm at a 55-millisecond time constant, then normalized and expressed as a percentage of the MVIC. The peak EMG for each stance exercise was recorded from seconds 2 through 7 of the 10-second total recording. The peak EMG recorded during the series of

3 squats was the EMG value used for analysis. Statistical analyses were performed using SPSS for windows, version 15.0 (SPSS, Inc., Chicago, IL, USA). A repeated measures ANOVA was conducted to examine differences in EMG activity of the gluteus medius during all exercises. Post hoc Bonferroni adjusted *t*-tests were performed to analyze specific comparisons among the various exercises. All tests were conducted with $\alpha = 0.05$.

RESULTS

Figure 6 summarizes the peak gluteus medius activity as a mean percentage of the MVIC. Repeated measures ANOVA indicated a significant difference in EMG activation of the gluteus medius across exercises ($F=40.259, p < 0.001$). Post hoc Bonferroni adjusted *t*-tests revealed that all single limb exercises resulted in significantly greater gluteus medius activity compared with double limb stance. Furthermore, single limb squatting produced significantly greater gluteus medius EMG activity compared with single limb stance

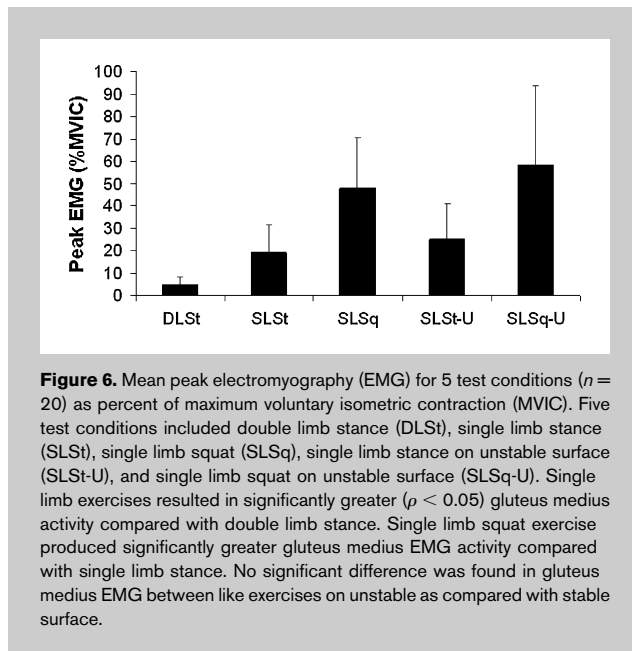


Figure 6. Mean peak electromyography (EMG) for 5 test conditions ($n = 20$) as percent of maximum voluntary isometric contraction (MVIC). Five test conditions included double limb stance (DLSt), single limb stance (SLSt), single limb squat (SLSq), single limb stance on unstable surface (SLSt-U), and single limb squat on unstable surface (SLSq-U). Single limb exercises resulted in significantly greater ($p < 0.05$) gluteus medius activity compared with double limb stance. Single limb squat exercise produced significantly greater gluteus medius EMG activity compared with single limb stance. No significant difference was found in gluteus medius EMG between like exercises on unstable as compared with stable surface.

($p < 0.001$). The EMG recording did not demonstrate significantly greater gluteus medius activation during like exercises on unstable as compared with stable surfaces with either the single limb stance ($p = 0.327$) or the single limb squat exercise ($p = 0.495$).

DISCUSSION

We hypothesized that single limb exercises would require greater gluteus medius activity than double limb stance. Our results confirmed this hypothesis. Our second hypothesis was that dynamic activities would require greater activation of the gluteus medius than static exercise. Our results also confirmed single limb squat produced greater gluteus medius EMG activity than single limb stance. Our third hypothesis was an exercise on an unstable surface would require greater gluteus medius activity than the same exercise on a firm surface. Although there was a trend toward an increased level of EMG activity on the unstable surface, the activation of the gluteus medius was not statistically greater than EMG activity of the same exercise performed on a stable surface.

The values obtained in this study add to the current body of research of gluteus medius activation during weight-bearing exercises. Schmitz et. al. (15) examined whether gluteus medius activity increased in response to isometric, sub-maximal, closed-chain external hip rotation forces in various positions of hip and knee flexion using surface EMG. The authors used a pulley-belt system to apply different levels of posterior directed forces to the lateral pelvis on the nonstance side. Their results indicated that EMG activity of the gluteus medius increased in relation to the applied force. Forward movement of the upper body relative to the base of support decreased gluteus medius EMG activity. Although the authors measured gluteus medius activity during a closed

kinetic chain activity, they only did so isometrically and did not normalize the EMG data. Thus, comparison between subjects could not be made.

In 2005, Earl (4) conducted a study similar to that of Schmitz et. al (15) but considered combined hip abduction, external rotation, and internal rotation forces. The subjects in this study were positioned in single limb stance while a cable system provided an abduction-internal rotation force, a straight abduction force, or an abduction-internal force. Recording surface EMG activity from the anterior and middle portions of the gluteus medius, the authors found significantly more muscle activation in both sections of the gluteus medius as the load increased. They also found that the EMG activity that was the greatest for both the anterior and middle gluteus medius occurred with the abduction and internal rotation exercise. Again, the authors did not normalize the data, again making intersubject comparisons difficult.

Our research protocol incorporated EMG measurements during a single leg stance; however, we did not include additional external forces such as a pulley or cable system. We used body weight resistance because we believed this to be typical of exercises commonly prescribed to strengthen the gluteus medius. We sought to increase demand on the gluteus medius by altering the surface condition. Although not reaching a level of significance, the increase in EMG suggests a trend toward increased recruitment of the gluteus medius when a single limb exercise is performed on an unstable surface.

Bolgla and Uhl (3) recorded surface EMG activity of the right gluteus medius in 3 non-weight-bearing exercises (sidelying hip abduction, standing hip abduction of the right extremity with a light cuff weight with the hip flexed 0° and 20°) and 3 weight-bearing exercises (standing hip abduction of the left extremity with a light cuff weight with the right hip flexed 0° and 20° and left-sided pelvic drop). After normalizing the EMG to an MVIC, the authors were able to rank the exercises according to the level of gluteus medius activation. Weight-bearing exercises demonstrated significantly greater EMG activity than the standing non-weight-bearing hip abduction exercises. One limitation acknowledged by the authors was the variability in trunk position during the exercises. Although subjects were told to keep their trunk vertical, this was not objectively monitored. This may influence results because of the possibility of inconsistent imposed loads on the hip abductors. In our study, we used a plumb line to monitor trunk position during the exercises to limit the influence of variable trunk positions.

More recently, Ayotte et al. (1) quantified the muscle activation patterns of the gluteus medius with surface electrodes during 5 unilateral weight-bearing exercises including the wall squat, a mini squat, and a forward step-up, lateral step-up, and retro step-up. After normalizing the EMG to an MVIC, the authors found the wall squat resulted in significantly greater gluteus medius activation than the

mini squat, lateral step-up, and retro step-up exercises. In descending order, gluteus medius activity as a percent of MVIC was as follows: wall squat, front squat, lateral step-up, retro step-up, and, finally, unilateral mini-squat. The authors concluded both the wall squat and the front step-up elicited sufficient mean gluteus medius EMG signal amplitude to provide an adequate strengthening stimulus. Different from our study, all exercises in this study were performed on a stable surface.

When analyzing EMG values, one should realize that the values represent levels of motor unit recruitment. As greater demands are imposed on a muscle, motor unit recruitment likewise increases. We found the single leg squat exercises on an unstable surface elicited the greatest level of EMG activity, implying that dynamic, unilateral exercises performed on an unstable surface impose greater demand on the gluteus medius. Although the EMG values represent motor recruitment, it is important to note they do not directly represent the strength or force-producing capability of a muscle. In addition, several factors may influence EMG values. One of these factors is the speed of contraction. The EMG values will be greater for higher-velocity contractions (2). We attempted to minimize the influence of velocity by monitoring the speed of movement with the squat exercises.

Limitations of our study include our sample group. Our young, healthy cohort limits external validity because the results may not be appropriately applied to individuals who have hip pain or pathology. In addition, although there was a trend toward greater EMG activity when exercises were performed on the unstable surface, this did not reach a level of significance. This may represent a type II error.

The results of this study provide the clinician objective measures of gluteus medius activation during various weight-bearing exercises. Single limb stance places more demands on the gluteus medius than double limb stance. Single limb squatting places more demands on the gluteus medius than single limb stance. Although the EMG values did not show a significant difference in gluteus medius activation when exercises were performed on a stable versus unstable surface, they may have clinical relevance as well as proprioceptive and functional implications.

PRACTICAL APPLICATIONS

Our results provide clinicians objective information describing the activation of the gluteus medius during various exercises. The clinician can implement specific exercises based on the amount of gluteus medius activation desired, progressing patients from static exercises requiring less activation to more challenging dynamic exercises requiring more gluteus medius activation. Our results suggest that if the intent is to provide a greater challenge to the gluteus medius, weight-bearing exercises should be performed unilaterally. Although we did not find the surface condition statistically significant, single limb dynamic exercise on an unstable

surface, such as an Airex cushion, may have clinical relevance, imposing greater demands on the muscle as indicated by the greater levels of EMG activity.

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